



COURSE SLIDES

# AI Agents for Stock Trading (C177)

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Course Code: C177

Conducted by Tertiary Infotech Academy Pte Ltd · UEN 201200696W

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COURSE ADMINISTRATION

# Welcome & Housekeeping

# About the Trainer



## Course Trainer

Principal Trainer  
Tertiary Infotech Academy Pte. Ltd.

### ROLE

Principal Trainer, Tertiary Infotech Academy Pte. Ltd.

### CERTIFICATION

Applied AI, quantitative research, and automation practitioner

### DELIVERS

bounded AI agents, market-data validation, strategy research, backtesting, risk controls, and paper trading

### FOUNDER

Founder and lead instructor at Tertiary Infotech / Tertiary Courses.

# Let's Know Each Other

- 1 Your name and one trading-research task that you would like an AI agent to help with.
- 2 One trading decision or action you would never allow an AI model to perform without human review.
- 3 One source of evidence you would require before trusting a trading claim.

# Ground Rules

1

Set your mobile phone to silent mode.

2

Participate actively — no question is too small.

3

Mutual respect: agree to disagree.

4

One conversation at a time.

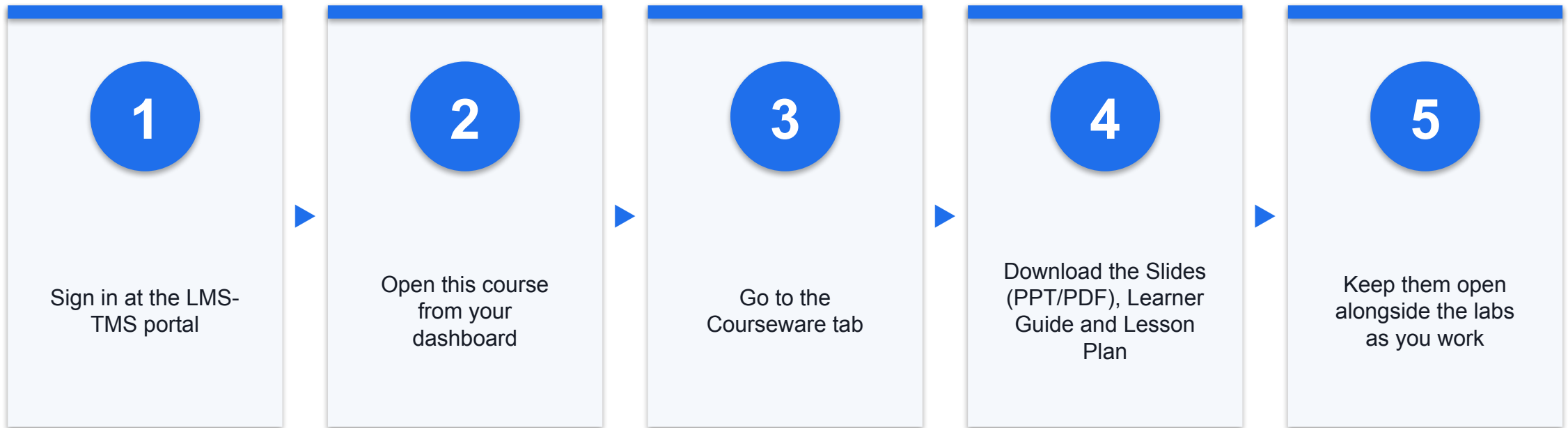
5

Be punctual; return from breaks on time.

6

Learn by doing — try every hands-on lab.

# Download Course Material



**Portal:** <https://lms-tms.tertiaryinfotech.com>

# Lesson Plan — 1 Day, 8 Scheduled Hours

### Course flow

- Day 1 — From Evidence-Led Research to a Controlled Paper Trade
  - Topic 1: From Trading Idea to Quality Market Data (Labs 1–2)
  - Topic 2: The Six-Step Trading Workflow and Staying in Control (Labs 3–4)

### Timing & approach

- Daily timing
  - 9:00 am - 6:00 pm (1-hour lunch; two 15-minute tea breaks)
- Learning arc
  - Concept → guided build → verify → discuss → recap
- Connected practice
  - Four labs build one bounded Northstar SPY research pack from hypothesis to controlled paper evidence.

# Learning Outcomes

1

## Bounded Agent

Explain how a bounded AI trading agent combines a model, instructions, tools, state, evidence, guardrails, and human control without becoming an autonomous money manager.

2

## Testable Hypothesis

Research a trading idea with an AI agent and convert it into a structured, falsifiable hypothesis with explicit rules, assumptions, evidence, and rejection criteria.

3

## Quality Market Data

Connect the agent workflow to an authorised market-data source and validate coverage, timestamps, corporate-action adjustments, completeness, consistency, and freshness before use.

4

## Backtest and Audit

Define deterministic trading rules, run an honest backtest, compare a benchmark, and audit data leakage, friction assumptions, overfitting, and out-of-sample behaviour.

5

## Risk-Sized Ticket

Calculate a bounded position size from account risk and stop distance, then prepare an auditable paper-order ticket with pre-trade limits and human approval evidence.

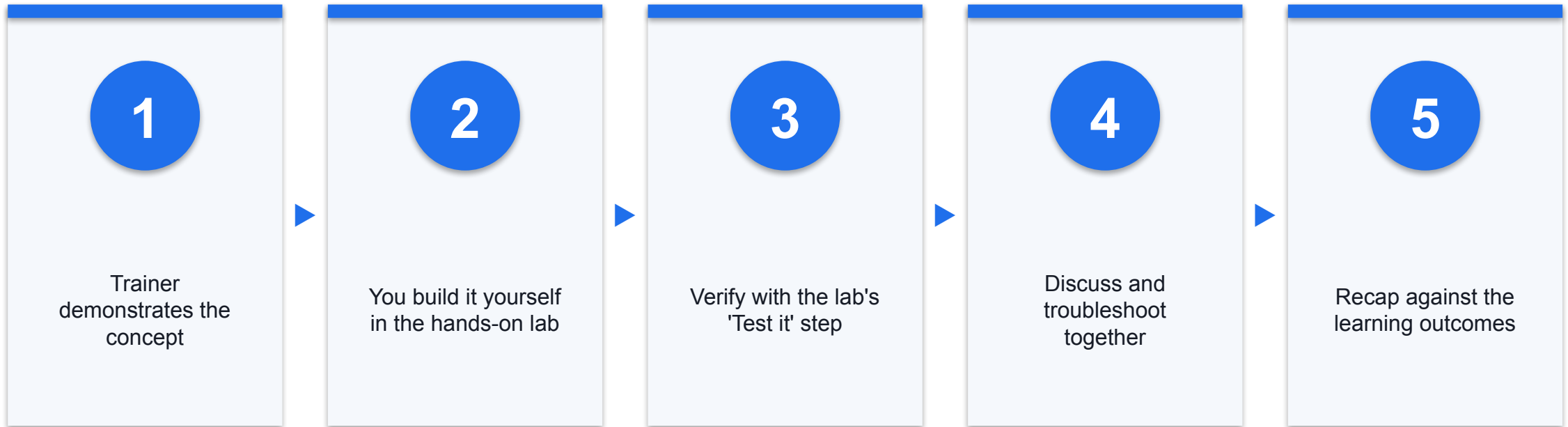
6

## Controlled Paper Trade

Place and verify a paper-only trade through a controlled agent workflow while applying verify-then-trust checkpoints, logs, stop conditions, and rollback procedures.

# How You'll Learn

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CORE CONCEPTS

# Course Fundamentals

# The Evidence-Led Trading Agent

1

## Research, not prediction

The agent turns an idea into testable evidence; it does not promise a market outcome.

2

## Tools, not hidden arithmetic

Typed Python tools fetch, validate, backtest, size, and submit only within code-enforced bounds.

3

## Paper, not live

Every order path is hard-wired to simulation and remains behind an explicit human gate.

4

## Artifacts, not chat memory

Versioned JSON, CSV, Markdown, plots, fingerprints, and receipts form the audit trail.

# The Six-Step Verify-Then-Trust Workflow

1

## Define

Freeze the HYPER hypothesis and deterministic rules.

2

## Validate

Accept only fit-for-purpose market data with recorded lineage.

3

## Backtest

Use causal timing, visible frictions, a benchmark, and a holdout.

4

## Audit

Check leakage, trials, stability, limitations, and decision criteria.

5

## Size

Compute quantity from human-set risk and exposure limits.

6

## Paper trade

Approve, submit to simulation, reconcile, verify, and roll back.

## COURSE PRINCIPLE

# A confident answer is not trading evidence.

Trust only the parts you can trace to a source, deterministic calculation, recorded decision, or verified paper receipt.

# What You'll Build

## Research Pack

- Structured HYPER hypothesis
- Cited research log
- Schema and version checks

## Evidence Pack

- Market-data contract
- Quality report and fingerprint
- Backtest, trades, chart, and audit

## Paper-Trade Pack

- Bounded risk calculation
- Dry-run ticket and approval
- Paper receipt and rollback record

# How Every Lab Progresses

1

Start from a verified checkpoint.

2

Use the agent only within its evidence boundary.

3

Run deterministic checks before interpretation.

4

Save an observable artifact and fingerprint.

5

Stop, resolve, or continue through a named gate.

TOPIC 01

# From Trading Idea to Quality Market Data

AI trading agents | research evidence | structured hypotheses | market-data connections | data-quality gates

# Key Concepts — From Trading Idea to Quality Market Data

1

### Bounded agency

The model may research and recommend, but deterministic tools calculate, controls constrain, and a person owns consequential decisions.

2

### Falsifiable hypothesis

A useful idea states the asset, data, signal, entry, exit, horizon, benchmark, costs, risks, and rejection rule before testing.

3

### Evidence chain

Every material claim links to a source, timestamp, tool result, assumption, or named owner instead of model confidence.

4

### Market-data contract

The workflow records provider, feed, symbol, timezone, session, adjustment, interval, coverage, and intended use.

5

### Quality gates

Schema, chronology, uniqueness, completeness, OHLC consistency, volume, freshness, and corporate actions are checked before analysis.

6

### Human control

The agent stops when evidence is missing, conflicts remain, data quality is weak, or an action exceeds paper-only authority.

# Chatbot, Workflow, Agent, or Autopilot?

## Bounded research system

- Chat explains; workflow follows fixed rules
- Agent chooses among narrow read-only tools
- Code calculates and validates deterministically
- A person approves any paper order

## Unsafe autopilot claim

- Model invents or changes trading rules mid-run
- One fluent answer is treated as evidence
- Keys or live-order endpoints are exposed
- No stop condition, audit trail, or owner exists

# Seven Building Blocks of a Bounded Trading Agent

1

## Goal

Turn one idea into a traceable research pack, not a buy or sell promise.

2

## Model

Interprets questions, plans tool use, and drafts explanations; it is not the numeric authority.

3

## Instructions

Define evidence, output schema, paper-only boundary, stop rules, and escalation.

4

## Tools

Read market data, validate it, run a backtest, size risk, and prepare or submit a paper ticket.

## Seven Building Blocks of a Bounded Trading Agent (cont.)

5

### State

Versioned JSON, CSV, and logs carry verified facts between labs.

6

### Guardrails

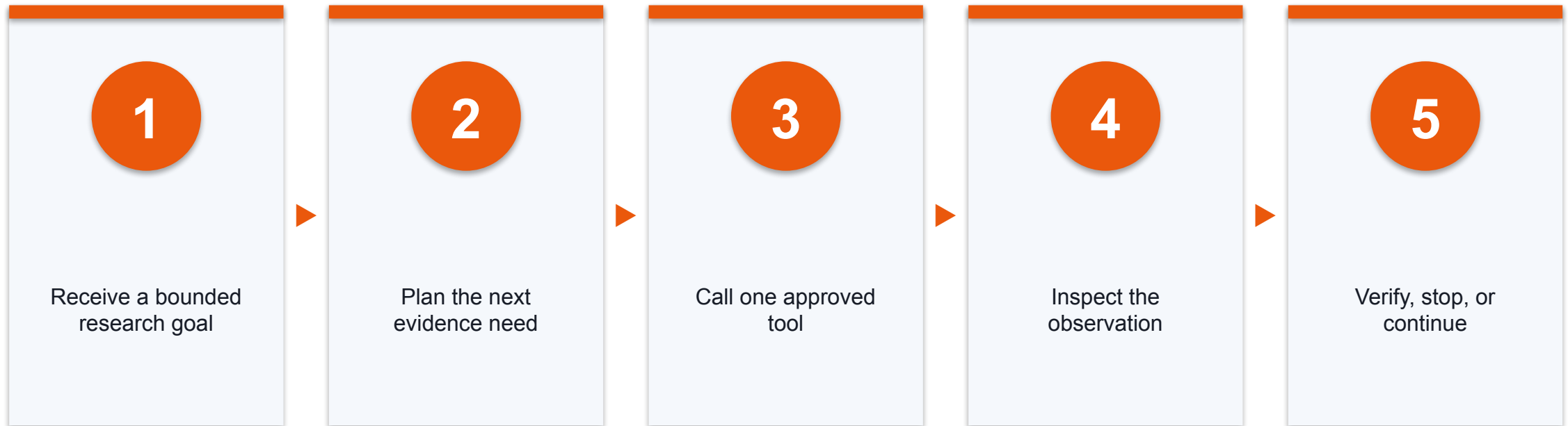
Allowlisted symbols, bounded dates, quantity caps, paper endpoint, dry run, and approval phrase.

7

### Human owner

Reviews exceptions, approves the paper ticket, and decides whether research continues.

# The Research Agent Loop



# A Trading Idea Is Not Yet a Hypothesis

## Vague idea

- Buy when momentum looks strong
- Use a good stock and recent data
- Exit when conditions change
- The strategy should beat the market

## Structured hypothesis

- SPY daily adjusted bars, fixed date range
- 20-day SMA crosses above 50-day SMA after close
- Long next session; exit on opposite cross
- Compare with buy-and-hold after stated frictions

# The HYPER Hypothesis Contract

1

## H - Hypothesis

One falsifiable sentence and the market rationale it is intended to test.

2

## Y - Yardstick

Benchmark, metrics, minimum sample, and rejection criteria defined in advance.

3

## P - Parameters

Asset, interval, lookbacks, entry, exit, position state, and date range.

4

## E - Evidence

Source URLs, market-data contract, assumptions, and OWNER\_TO\_VERIFY items.

5

## R - Risks

Look-ahead, overfitting, costs, liquidity, regime change, model error, and operational failure.

## Research Evidence: Stronger and Weaker Uses

### Stronger evidence practice

- Exchange, regulator, provider, or official SDK documentation
- Direct dataset metadata and timestamped tool output
- Independent source used to cross-check a material claim
- Citation, access date, scope, and limitation recorded

### Weaker evidence practice

- Uncited model memory or confident prose
- Screenshot with no source, time, or feed
- Promotional performance claim treated as proof
- Later edits that overwrite the original reasoning

# An Agent Output Needs a Contract

1

## Claim

What is being proposed, in one falsifiable sentence.

2

## Known evidence

Source-backed facts and exact tool observations.

3

## Assumptions

Choices that shape the result but are not measured facts.

4

## Unknowns

Missing or conflicting items labelled OWNER\_TO\_VERIFY.

5

## Next tool

The single approved action needed to reduce uncertainty.

6

## Stop reason

Why the run cannot safely continue, when applicable.

# What a Daily OHLCV Bar Represents

1

## Timestamp

The labelled interval and timezone; it must match the intended trading session.

2

## Open

First eligible trade price represented by the bar's aggregation rules.

3

## High

Highest represented trade price; it should not be below open or close.

4

## Low

Lowest represented trade price; it should not be above open or close.

5

## Close

Final represented trade price; adjusted and raw closes answer different questions.

6

## Volume

Aggregated eligible share volume; feed coverage affects the observed value.

# Feed Coverage Changes What You Observe

## Single-exchange feed

- Lower-cost access and useful for initial app testing
- Only trades and quotes from the named venue
- Volume and intraday prices may differ from the wider market
- Scope must be recorded in the data contract

## Consolidated feed

- Combines eligible activity across US exchanges
- Broader price, quote, and volume representation
- May require a paid entitlement or provider plan
- Still requires timezone, session, and adjustment checks

# Time, Session, and Signal Availability

## Known at decision time

- Completed bars with unambiguous timezone
- Signal computed after the bar closes
- Order scheduled for the next eligible session
- Calendar and session assumptions recorded

## Known only later

- Current bar's final close before it occurs
- Next session open used to decide today's signal
- Revised data silently substituted after the run
- Mixed local, exchange, and UTC dates

# Corporate Actions and Adjusted Prices

1

## Split

Share count and quoted price change mechanically; an unadjusted series can show a false crash.

2

## Dividend

Cash distribution affects total return even when the price series alone does not show it.

3

## Adjustment mode

Raw, split-adjusted, or fully adjusted data must be named in the contract.

4

## Benchmark consistency

Strategy and benchmark must use comparable return and adjustment assumptions.

5

## Point-in-time concern

Current symbol mappings and revised histories can leak later knowledge into old dates.

6

## Verification

Inspect provider metadata and a known event before accepting the series.

# The Market-Data Contract

1

## Provider and endpoint

Who supplied the data and which documented route returned it.

2

## Feed and entitlement

IEX, SIP, delayed, or another feed plus the account's access scope.

3

## Instrument

Symbol, asset type, venue context, and currency.

4

## Time definition

Interval, start, end, timezone, session, and as-of date.

5

## Transformations

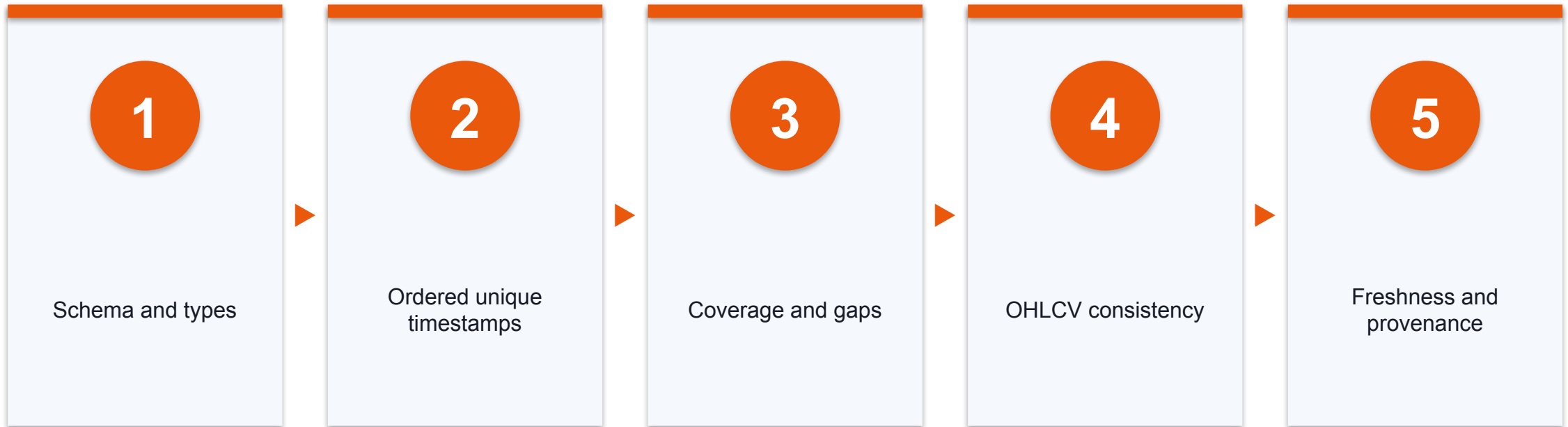
Adjustment, filtering, resampling, missing-value handling, and version.

6

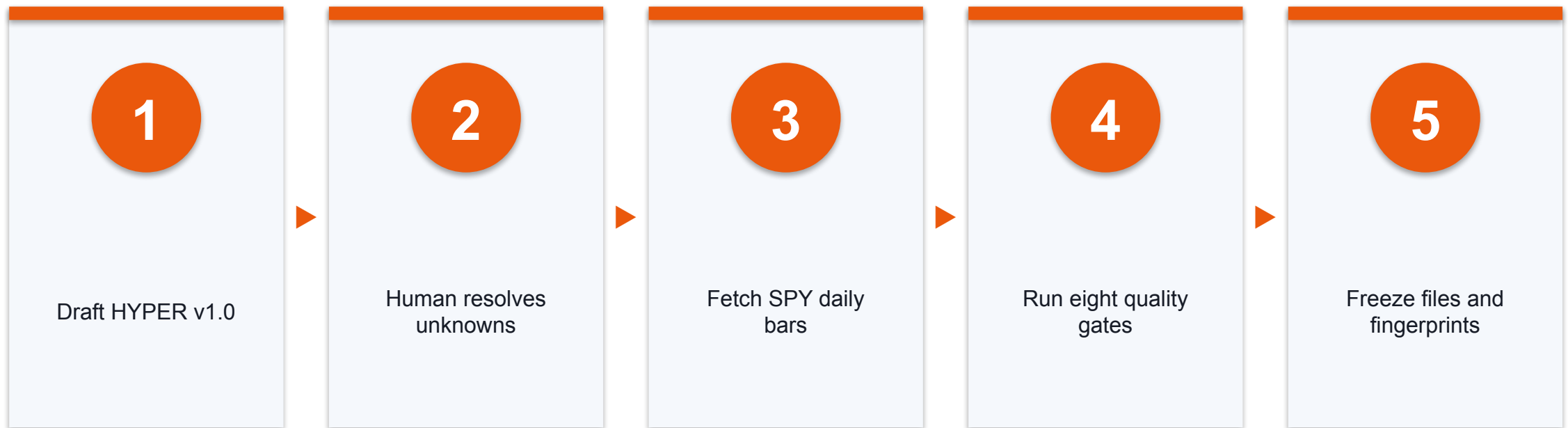
## Intended use

Educational daily-bar research, not a live executable-price guarantee.

# Eight Data-Quality Gates



## Worked Example: Freeze the SPY Research Pack



# Hands-On Labs — From Trading Idea to Quality Market Data

## Lab 1

- Research an Idea and Freeze a Testable Hypothesis

## Lab 2

- Fetch Market Data and Prove It Is Fit for Purpose

# Research an Idea and Freeze a Testable Hypothesis

## LAB 1

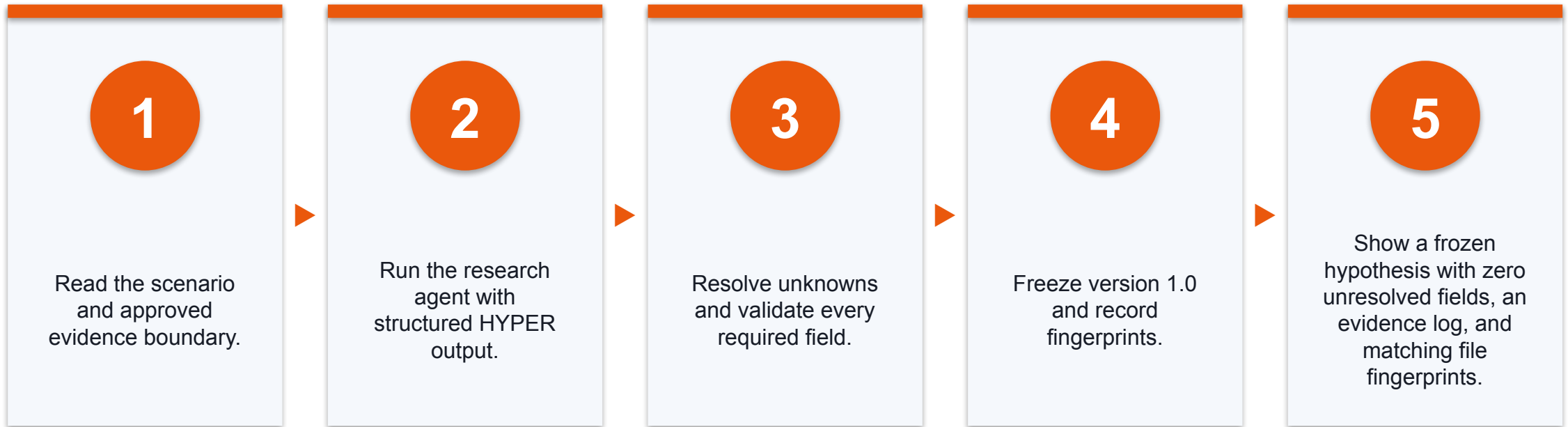
Start the connected SPY research scenario by running an OpenAI Agents SDK research agent with read-only course-source tools and structured output. Separate evidence, assumptions, and unknowns; resolve the required human-owned fields; validate the schema; then freeze version 1.0 with fingerprints before any market data is fetched.

### You'll build

**C177-trading-agent-pack/01-hypothesis/** containing **hypothesis\_spec.json**, **research\_log.md**, **agent\_run\_manifest.json**, **validation\_report.json**, and **fingerprints.json** for the frozen SPY 20/50-day moving-average hypothesis

**Resources:** Use the matching lab's prerequisites and supplied files.

# Lab 1 — Key Route



# Fetch Market Data and Prove It Is Fit for Purpose

## LAB 2

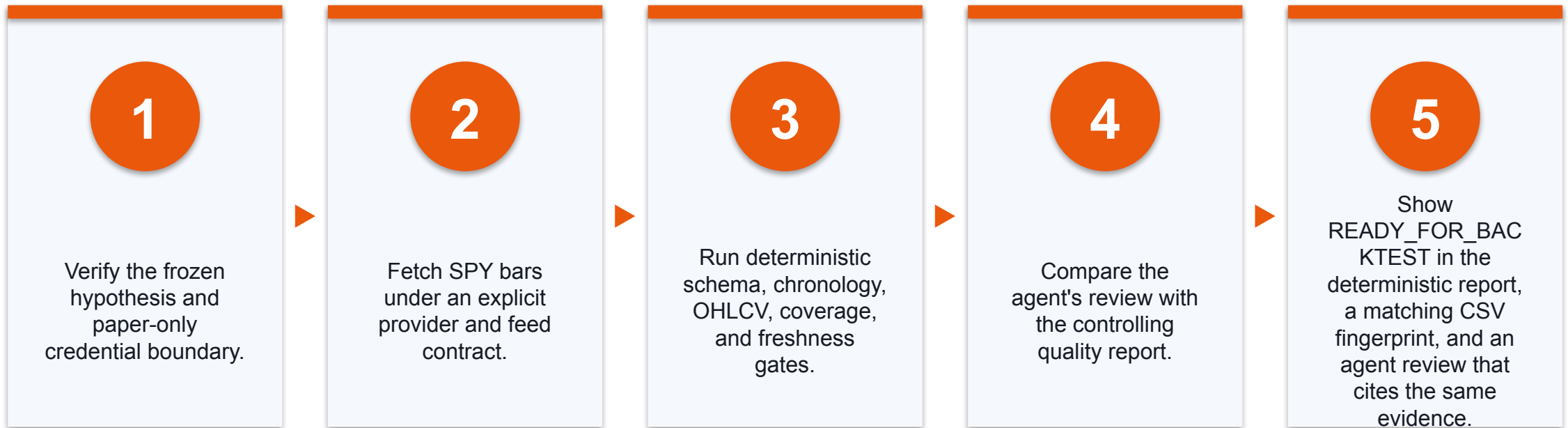
Verify the frozen hypothesis, connect to Alpaca's historical stock-data client, fetch the declared SPY daily bars, and produce a source contract, CSV, quality report, and fingerprint. Then give the research agent read-only access to the structured quality report so it can recommend READY\_FOR\_BACKTEST or HOLD\_DATA without changing any numeric observation.

### You'll build

**C177-trading-agent-pack/02-market-data/** containing `spy_daily.csv`, `market_data_contract.json`, `data_quality_report.json`, `data_agent_review.json`, `retrieval_manifest.json`, and `fingerprints.json`

**Resources:** Use the matching lab's prerequisites and supplied files.

## Lab 2 — Key Route



## Recap — From Trading Idea to Quality Market Data

1

You can now: LO1 and LO2: operate a bounded tool-using research agent and convert one trading idea into a cited, structured, falsifiable HYPER hypothesis

2

You can now: LO3: connect the workflow to authorised Alpaca market data, record the data contract, run deterministic quality gates, and let the agent interpret only the verified report

TOPIC 02

# The Six-Step Trading Workflow and Staying in Control

rule specification | honest backtesting | audit checks | risk sizing | paper orders | verify-then-trust checkpoints

## KEY CONCEPTS

# Key Concepts — The Six-Step Trading Workflow and Staying in Control

1

### Rules before results

Entry, exit, timing, sizing, costs, benchmark, and rejection criteria are frozen before the engine runs.

2

### Deterministic core

Code owns calculations and enforcement; the model interprets, organises evidence, and explains uncertainty.

3

### Backtest realism

Signals are shifted, data is split, and fees, spread, slippage, fills, and benchmark assumptions are visible.

4

### Risk budget

Quantity follows a capped account-risk budget and stop distance, never a model's confidence score.

5

### Paper-only execution

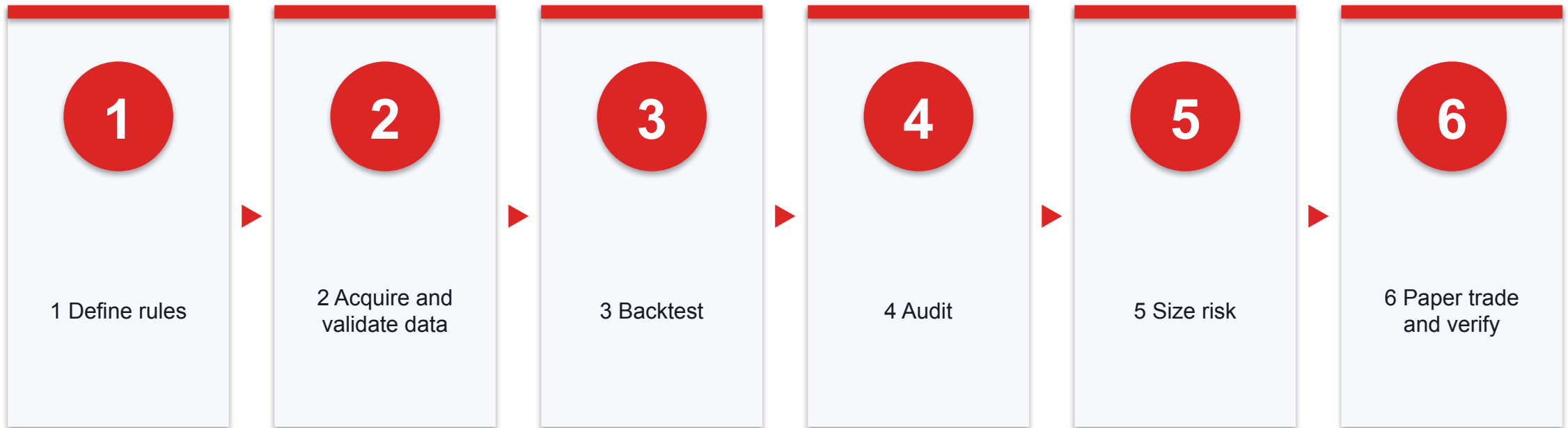
The supplied client uses the paper endpoint, a symbol allowlist, dry run, quantity cap, and explicit approval token.

6

### Verify then trust

Every stage produces checks, logs, fingerprints, a decision, and a safe rejoin point.

# The Six-Step Controlled Trading Workflow



## Rule Specification: Remove Every Hidden Choice

1

### Universe

One named asset or a point-in-time selection rule.

2

### Signal

Exact formula, lookback, bar interval, and data field.

3

### Timing

When the signal becomes known and when the order is assumed to execute.

4

### Entry and exit

Conditions, position state, and conflict resolution.

## Rule Specification: Remove Every Hidden Choice (cont.)

5

### Sizing

Fixed rule independent of model enthusiasm.

6

### Friction

Fees, spread or slippage, and fill assumptions.

7

### Benchmark

A comparable alternative such as buy-and-hold.

8

### Decision

Predeclared evidence for CONTINUE, HOLD, or REVISE.

# Look-Ahead Bias: The One-Bar Test

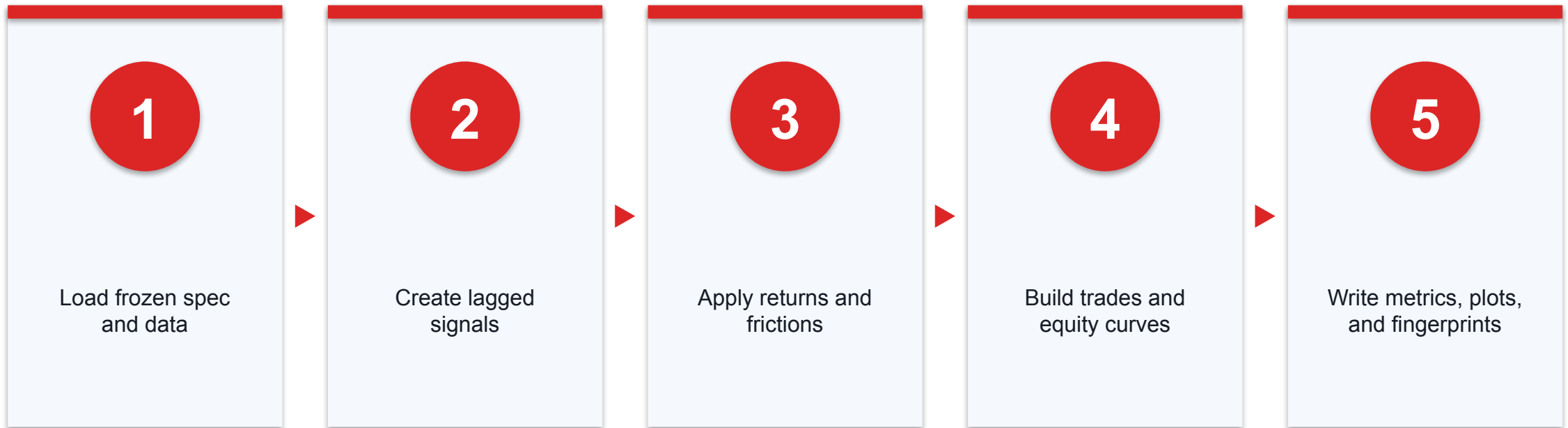
## Biased implementation

- Today's close creates today's signal
- Today's signal receives today's close-to-close return
- Final high or low is used before the bar completes
- Future revisions influence an earlier decision

## Causal implementation

- Completed bar creates the signal after close
- Signal is shifted before the next return is applied
- Execution timing is written in the rule contract
- Inputs and code versions are fingerprinted

# Backtest Engine: Inputs to Evidence



## Development Window and Out-of-Sample Window

### Development segment

- Used to implement and debug the fixed rule
- Can reveal obvious data or code errors
- Results may influence design choices
- Must not be presented as untouched evidence

### Out-of-sample segment

- Held back until the specification is frozen
- Used once for a cleaner generalisation check
- Compared with benchmark and decision criteria
- Poor evidence leads to HOLD, not repeated tuning

# Read Metrics as a System, Not a Score

1

## Total return

Growth over the window; it depends strongly on start and end dates.

2

## Annualised return

Compounded rate normalised for time; unstable with short samples.

3

## Volatility

Dispersion of returns, not a complete measure of loss risk.

4

## Sharpe ratio

Excess return per unit of volatility under stated assumptions.

## Read Metrics as a System, Not a Score (cont.)

5

### Maximum drawdown

Largest peak-to-trough decline in the simulated equity curve.

6

### Turnover

How often exposure changes; it drives friction sensitivity.

7

### Trade count

A tiny sample cannot support a stable hit-rate story.

8

### Benchmark gap

Strategy result minus a comparable passive alternative.

## Frictionless Results and More Realistic Results

### Frictionless simplification

- Every order fills immediately at the chosen bar price
- No spread, fee, slippage, latency, or market impact
- Unlimited liquidity at any quantity
- Useful only as an upper-bound diagnostic

### Declared reality assumptions

- Turnover incurs stated fee and slippage basis points
- Execution timing and price field are explicit
- Paper limitations and liquidity are disclosed
- Sensitivity cases show how conclusions change

# Overfitting Red Flags

1

## Many trials

Dozens of unreported variants make the best result look rarer than it is.

2

## Tiny sample

Few trades or one market regime cannot support a stable conclusion.

3

## Sharp optimum

Only one exact parameter works while nearby values collapse.

4

## No benchmark

A complex strategy may merely reproduce market exposure.

5

## Hidden exclusions

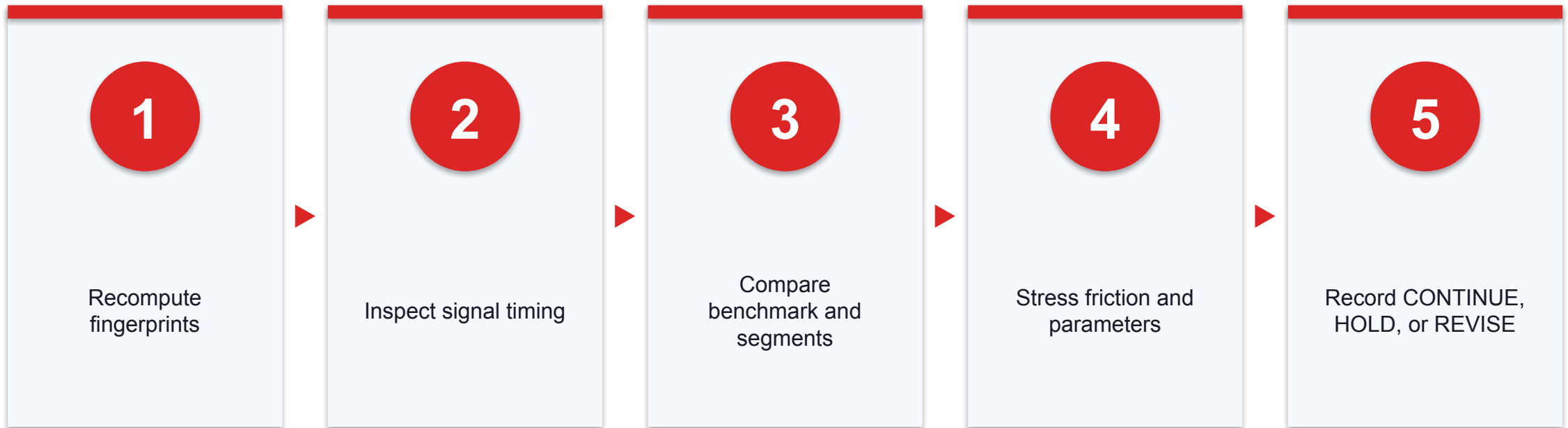
Changing assets or dates after inspection creates selection bias.

6

## Holdout erosion

Repeatedly tuning on the same out-of-sample period destroys its role.

# The Backtest Audit Trail



# Model Work and Deterministic Work

## AI model may

- Summarise sources and organise a hypothesis draft
- Choose the next approved read-only tool
- Explain metrics and surface uncertainty
- Draft a HOLD or CONTINUE rationale for review

## Code or human must

- Validate schema, timestamps, prices, and fingerprints
- Compute signals, returns, costs, and position size
- Enforce paper endpoint, allowlist, caps, and dry run
- Approve the ticket and own the decision

## Position Size Comes from Risk Budget and Stop Distance

1

### Account equity

The paper account value used only for the exercise.

2

### Risk fraction

A small predeclared fraction chosen by the human owner, not the agent.

3

### Risk budget

Account equity multiplied by risk fraction.

4

### Per-share risk

Absolute difference between planned entry and protective stop.

5

### Risk quantity

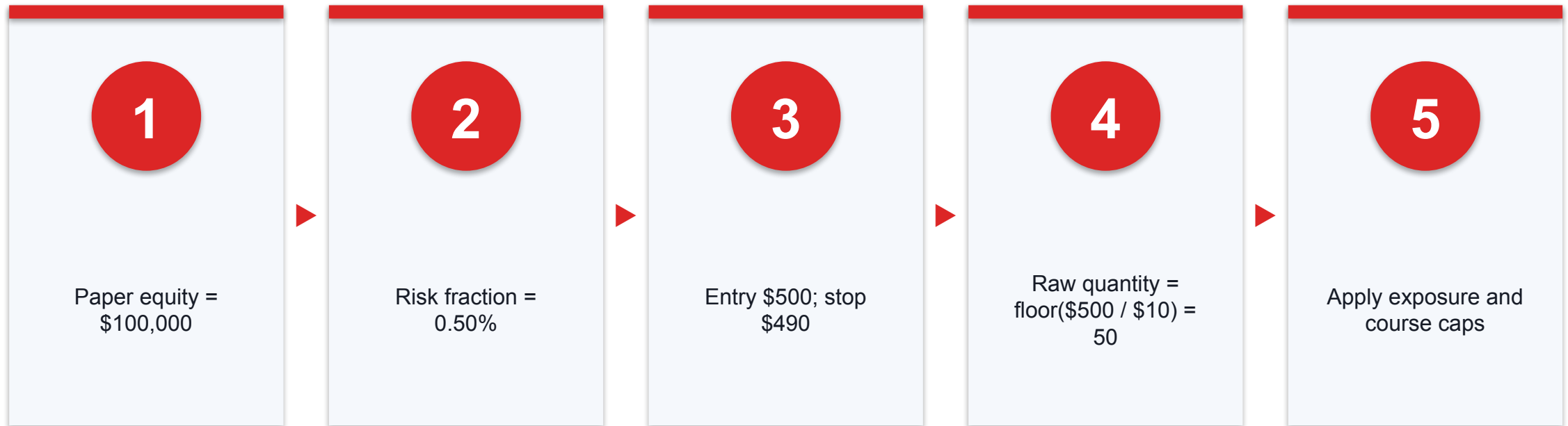
Floor of risk budget divided by per-share risk.

6

### Exposure cap

A second ceiling based on maximum position value and available buying power.

## Worked Position-Size Example



# The Pre-Trade Paper Ticket

1

## Instrument

Allowlisted symbol, asset type, side, and paper environment.

2

## Evidence

Accepted hypothesis, data, backtest, audit decision, and fingerprints.

3

## Quantity

Risk calculation, exposure cap, buying-power check, and hard maximum.

4

## Order

Type, time in force, reference price, and paper endpoint.

5

## Limits

No live route, no secret in logs, no unsupported symbol, no blank approval.

6

## Owner

Reviewer, approval time, approval token, and reason for the exercise.

# Paper Trading Is Useful and Incomplete

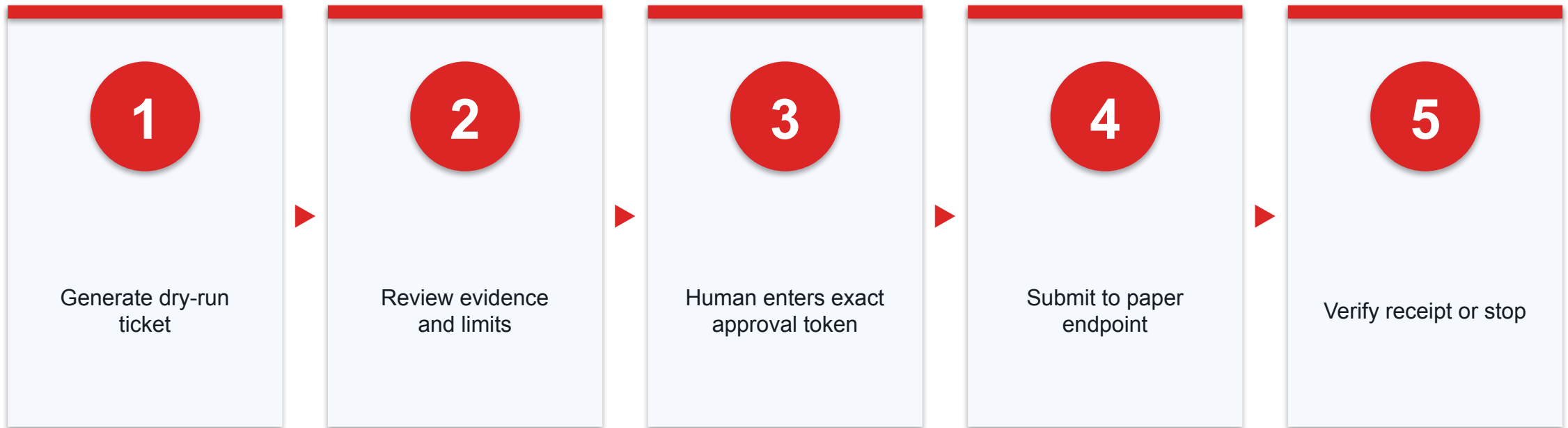
## What paper trading exercises

- Authentication, request schema, account checks, and order lifecycle
- Real-time data handling and software error paths
- Logging, approval, idempotency, and receipt verification
- Operational rehearsal without risking capital

## What it may not reproduce

- Market impact, information leakage, or latency slippage
- Queue position and realistic available liquidity
- All fees, dividends, price improvement, or live behaviour
- Future profitability or suitability for real money

# Human Gate and Safe Submission



# Verify-Then-Trust Checkpoints

1

## Hypothesis gate

Schema valid, assumptions separated, unknowns owned, version frozen.

2

## Data gate

Contract complete, quality checks recorded, fingerprint accepted.

3

## Backtest gate

Timing causal, benchmark present, segments and frictions reported.

4

## Audit gate

Leakage, trials, stability, limitations, and decision documented.

5

## Risk gate

Quantity recomputed from approved inputs and bounded by caps.

6

## Execution gate

Paper endpoint, human approval, unique ID, receipt, and reconciliation.

# Failure, Pause, Reconcile, and Roll Back

1

## Tool failure

Record the error; do not fabricate a result or continue from stale state.

2

## Data drift

Freeze the new dataset separately and rerun quality and backtest evidence.

3

## Duplicate risk

Query the client order ID before retrying a timed-out request.

4

## Unexpected fill

Stop the workflow, inspect the paper account, and preserve the receipt.

5

## Model change

Rerun stable prompts and tool-choice checks before accepting explanations.

6

## Rollback

Cancel open paper orders, disable submission, and restore the last verified version.

## Hands-On Labs — The Six-Step Trading Workflow and Staying in Control

### Lab 3

- Backtest the Rules and Audit the Evidence

### Lab 4

- Size Risk, Approve, and Verify a Paper Trade

# Backtest the Rules and Audit the Evidence

## LAB 3

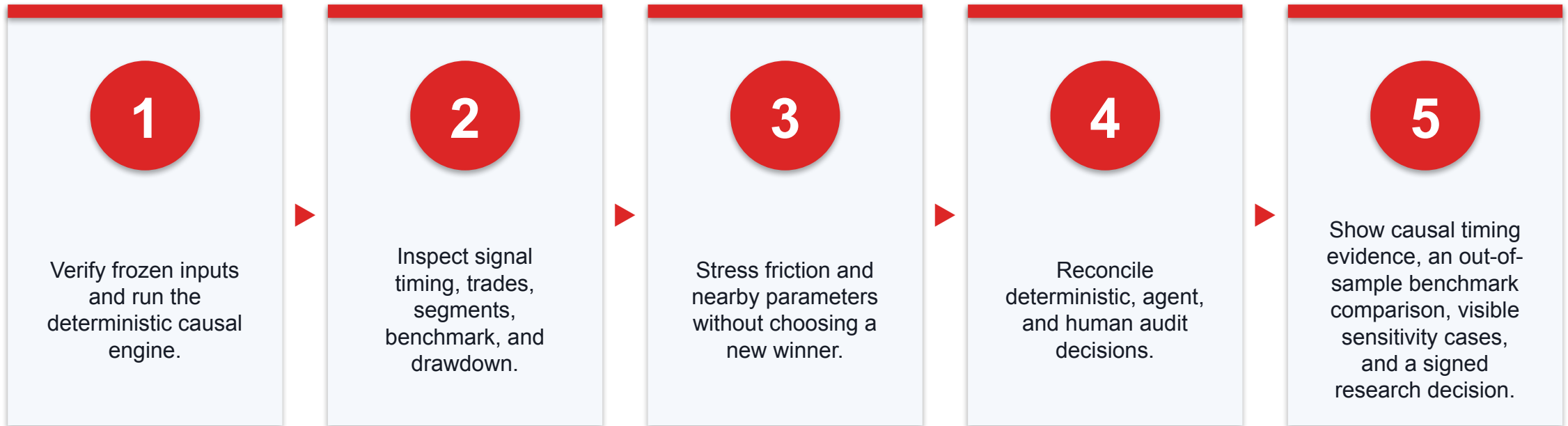
Verify the accepted hypothesis and market-data fingerprints, run the fixed 20/50-day crossover engine with a one-bar signal shift and declared frictions, then inspect trades, development and out-of-sample metrics, benchmark results, drawdown, and sensitivity checks. The agent receives only read-only report tools and drafts a review that the human auditor must reconcile with deterministic checks.

### You'll build

**C177-trading-agent-pack/03-backtest-audit/** containing **strategy\_spec.json**, **signal\_sample.csv**, **trades.csv**, **equity\_curve.csv**, **equity\_curve.png**, **backtest\_metrics.json**, **sensitivity\_report.json**, **audit\_log.md**, **backtest\_agent\_review.json**, **run\_manifest.json**, and **fingerprints.json**

**Resources:** Use the matching lab's prerequisites and supplied files.

## Lab 3 — Key Route



# Size Risk, Approve, and Verify a Paper Trade

## LAB 4

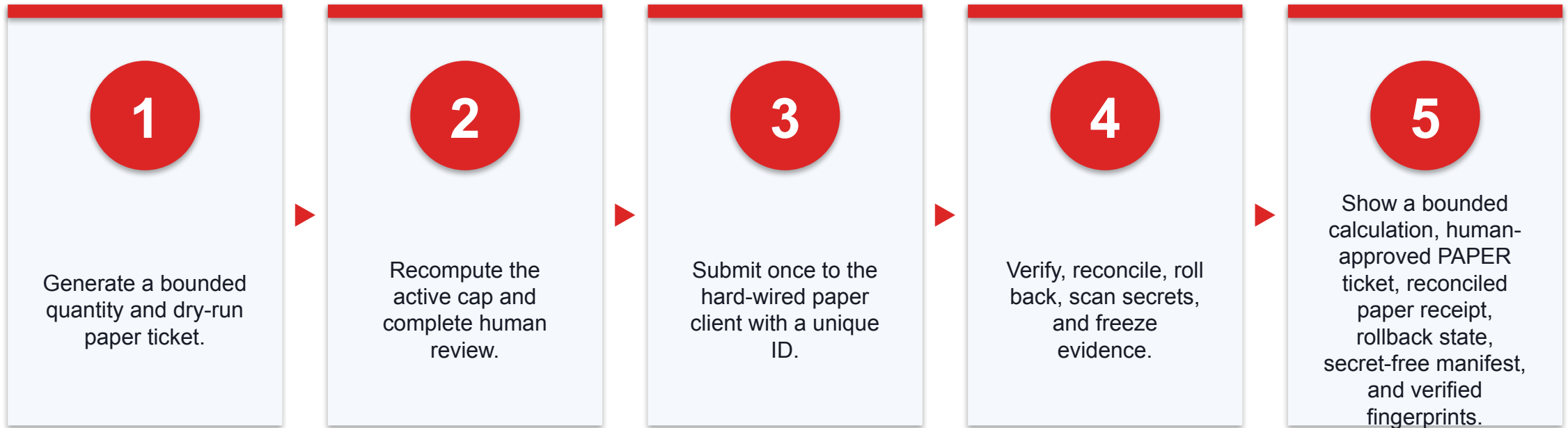
Use the accepted research evidence to calculate quantity from a human-set paper risk budget, stop distance, exposure ceiling, buying power, and hard course cap. Generate a dry-run ticket, let the agent perform a read-only evidence review, obtain explicit human approval, submit through an Alpaca TradingClient constructed with paper=True, reconcile the receipt, and close with a secret-free final manifest.

### You'll build

**C177-trading-agent-pack/04-paper-trade/** containing **risk\_calculation.json**, **paper\_order\_ticket.json**, **ticket\_agent\_review.json**, **human\_approval.md**, **submission\_attempt.json**, a paper receipt or structured error, **reconciliation\_log.md**, **rollback\_record.md**, **final\_manifest.json**, and **fingerprints.json**

**Resources:** Use the matching lab's prerequisites and supplied files.

## Lab 4 — Key Route



## Recap — The Six-Step Trading Workflow and Staying in Control

1

You can now: LO4: run a causal deterministic backtest, compare a benchmark and holdout, stress stated assumptions, and record an auditable research decision

2

You can now: LO5 and LO6: calculate a bounded quantity, generate and review a paper-order ticket, apply the human gate, submit only to simulation, and verify or roll back the result



WRAP-UP

# Course Summary & Next Steps

# What You Achieved

1

## Bounded Agent

Explain how a bounded AI trading agent combines a model, instructions, tools, state, evidence, guardrails, and human control without becoming an autonomous money manager.

2

## Testable Hypothesis

Research a trading idea with an AI agent and convert it into a structured, falsifiable hypothesis with explicit rules, assumptions, evidence, and rejection criteria.

3

## Quality Market Data

Connect the agent workflow to an authorised market-data source and validate coverage, timestamps, corporate-action adjustments, completeness, consistency, and freshness before use.

4

## Backtest and Audit

Define deterministic trading rules, run an honest backtest, compare a benchmark, and audit data leakage, friction assumptions, overfitting, and out-of-sample behaviour.

5

## Risk-Sized Ticket

Calculate a bounded position size from account risk and stop distance, then prepare an auditable paper-order ticket with pre-trade limits and human approval evidence.

6

## Controlled Paper Trade

Place and verify a paper-only trade through a controlled agent workflow while applying verify-then-trust checkpoints, logs, stop conditions, and rollback procedures.

## Continue with Evidence, Not Automation Pressure

1 Reproduce the same run before changing the idea.

2 Version every rule, dataset, prompt, code, and decision change.

3 Increase realism and independent review before increasing autonomy.

4 Keep execution paper-only throughout the course and personal practice.

# Keep Practising After the Course

1

Re-run the labs on your own machine — repetition is what makes the skills stick.

2

Apply the techniques to a real process or project in your own organisation.

3

Your course material stays available at <https://lms-tms.tertiaryinfotech.com/>.

4

Explore the follow-on courses at [www.tertiarycourses.com.sg](http://www.tertiarycourses.com.sg).

KEEP VERIFYING BEFORE YOU TRUST

# Thank You!

You can now direct a bounded AI agent through a traceable trading research cycle while keeping data, calculation, risk, and execution evidence under human control.